A finite-speed return bridge and its polygonal action

A velocity-resolved telegraph bridge gives one consistent random trajectory at every cut resolution. Its midpoint has a discrete probability mass, while the action error of sampled polygons vanishes at a controlled rate. The construction isolates velocity memory from the introduction of fresh noise at each cut.

1. Model and endpoint convention

Fix mass $m > 0$, speed $0 < u < c$, reversal rate $\lambda > 0$ and duration $T > 0$. In an inertial frame set $X_0 = 0$, $V_0 = u$ and

$$V_t = u(-1)^{N_t}, \qquad X_t = \int_0^t V_s \, ds,$$

where N is a homogeneous Poisson process of rate λ. Velocity is right-continuous. The rate is a supplied classical stochastic clock. This reduced model describes impulsive reversals; momentum receivers require the extra interaction bookkeeping of the physical-cut model (C030).

We construct the bridge to $(X_T, V_T) = (0, -u)$ by disintegrating switch-time densities at the interior position $X_T = 0$. The explicit simplex construction below fixes the version of this zero-probability conditioning. Before conditioning, the zero-switch event has probability $e^{-\lambda T}$ and lies at $(uT, +u)$; the chosen terminal velocity selects odd positive switch counts.

The free kinetic functional is $S[X] = (m/2) \int_0^T V_t^2 dt$, in action units. We compare it with the sampled polygon and with the zero-position chord. The chord matches position endpoints; its velocity endpoints differ from the bridge. All comparisons below name this positional endpoint convention.

2. Exact conditional path law

Write $N_T = 2K + 1$ and $z = \lambda T$. The bridge count law is

$$\boxed{\Pr(K = k \mid \text{bridge}) = w_k = \frac{(z/2)^{2k}}{(k!)^2 I_0(z)}, \qquad k = 0, 1, \ldots,}$$

where $I_0(z) = \sum_{k \geq 0} (z/2)^{2k}/(k!)^2$. Given $K = k$, independently choose uniform simplex vectors $(P_0, \ldots, P_k)$ and $(M_0, \ldots, M_k)$, each summing to $T/2$. Follow velocities $+u, -u, +u, -u, \ldots$ for durations $P_0, M_0, P_1, M_1, \ldots, P_k, M_k$. For $k = 0$ each simplex is a singleton. This specifies the entire conditioned path, including all later observations.

Derivation. Given $N_T = 2k + 1$, the ordered switch times are uniform in $0 < s_1 < \cdots < s_{2k+1} < T$. Their $2k + 2$ spacings are uniform on the simplex of total length T. The total positive duration A has density

$$f_k(a) = \frac{(2k+1)!}{(k!)^2 T^{2k+1}} a^k (T - a)^k, \qquad 0 < a < T.$$

Since $X_T = u(2A - T)$, the position Jacobian is $da/dx = 1/(2u)$. Multiplication by the Poisson count weight gives the joint density

$$\left. \frac{\Pr(X_T \in dx, N_T = 2k+1)}{dx} \right|_{x=0} = \frac{e^{-\lambda T} \lambda^{2k+1}}{2u(k!)^2} (T/2)^{2k}.$$

Its sum is $e^{-\lambda T} \lambda I_0(z)/(2u) > 0$. Division gives w_k. Conditioning the uniform spacings on $A = T/2$ leaves the two independent simplexes stated above. The occupation-time backbone is the equal-rate specialization of Cinque's Theorem 2.1, (2.6), printed p. 4; the source also gives the alternating switch representation (2.4), p. 3.

3. Midpoint law, including the atom

The exact midpoint law is an explicit pushforward of the preceding mixture. Let $d = (P_0, M_0, \ldots, P_k, M_k)$, $s_0 = 0$ and $s_j = \sum_{i<j} d_i$. Then

$$Y = X_{T/2} = u \sum_{j=0}^{2k+1} (-1)^j \min\{d_j, \max(0, T/2 - s_j)\}.$$

For any bounded measurable g, its expectation is $\sum_k w_k \mathbb{E}_k[g(Y)]$, where $\mathbb{E}_k$ is integration over the two normalized simplexes. This formula sums over midpoint velocities. The joint law retains $V_{T/2} = u(-1)^j$ on $s_j \le T/2 < s_{j+1}$.

In particular,

$$\boxed{\Pr(Y = uT/2) = \frac{1}{I_0(\lambda T)}.}$$

For $K = 0$ the unique switch occurs at $T/2$, and $Y = uT/2$ with right-continuous velocity $-u$. For $k \ge 1$, all simplex coordinates are strictly positive almost surely. Attaining $Y = uT/2$ would use all positive duration before the midpoint, leaving extra positive intervals with zero length. Attaining $Y = -uT/2$ would require zero initial positive duration. Both are simplex boundary events. The midpoint lies in an interval with index $1 \le j \le 2k$. For odd j its position is $u(2\sum_{i<j,\ i \text{ even}} d_i - T/2)$; for even j it is $u(T/2 - 2\sum_{i<j,\ i \text{ odd}} d_i)$. Each sum is a nonempty proper prefix of its simplex vector, so on each such region Y is a nonconstant affine function of the free simplex coordinates. Thus the $k \ge 1$ contribution is absolutely continuous on $(-uT/2, uT/2)$.

The velocity convention matters at the atom. On the one-switch component, conditioning instead on $|X_T| < \epsilon$ makes the switch time uniform in a symmetric interval around $T/2$. Half of these paths still have velocity $+u$ at the midpoint and half have $-u$. As $\epsilon \downarrow 0$, their position paths converge uniformly to the one-switch return path, but their midpoint velocity laws retain that equal mixture. Evaluation of velocity at a jump is discontinuous. Our exact bridge uses the right-continuous path version above; an endpoint-window measurement is a separate specified protocol.

4. Cuts sample one preparation

Every finite set of cuts is evaluated on this same count/simplex probability space. If ρ refines π, deleting the extra coordinates of $(X_t, V_t)_{t\in\rho}$ returns $(X_t, V_t)_{t\in\pi}$ pointwise. Hence their joint laws satisfy exact restriction consistency. This argument handles atoms without multiplying singular transition densities.

An independent midpoint-reset rule fails a direct test. Conditional on $Y = uT/2$, speed at most u forces velocity $+u$ almost everywhere on the first half and $-u$ almost everywhere on the second half. In particular $X_{T/4} = X_{3T/4} = uT/4$ deterministically. Giving either quarter point fresh nonzero variance changes the old preparation or violates its speed support. Velocity and the endpoint conditioning carry the required memory.

5. Exact action limit under arbitrary cuts

For any partition π of $[0, T]$, let X^π be the sampled linear interpolant, $|\pi|$ its largest interval, and $S_\pi = S[X^\pi]$. Then

$$S[X] = \frac{mu^2 T}{2}, \qquad 0 \le S[X] - S_\pi \le \frac{mu^2}{2} N_T |\pi|.$$

Indeed the polygon velocity on interval I is the average $\bar{V}_I$, and

$$S[X] - S_\pi = \frac{m}{2} \sum_{I \in \pi} \int_I (V_t - \bar{V}_I)^2 dt.$$

An interval without an interior switch contributes zero. Every other interval contributes at most $mu^2|I|/2$; their total length is at most $N_T|\pi|$. The count is finite almost surely. Differentiating the convergent positive series for I_0, with $I_1 = I_0'$, also gives

$$\mathbb{E}[N_T \mid \text{bridge}] = 1 + z\frac{I_1(z)}{I_0(z)}.$$

Consequently $S_\pi \to mu^2T/2$ almost surely and in L^1 on any deterministic shrinking meshes, with the displayed expected error bound. Also $\|X^\pi - X\|_\infty \leq 2u|\pi|$. Along nested partitions S_π increases, by square completion at each inserted node.

The surviving action relative to the zero-position chord is the kinetic cost of the fixed returning path. The polygon approximation error tends to zero. Its value $mu^2T/2$ tends to zero with duration or speed. These are separate limits from A01's long-observation stationary coefficient $H_* = mu^2/\lambda$. For the bridge midpoint observable, C031 gives $0 \leq \kappa_{\text{mid}} = 4m\operatorname{Var}(Y)/T \leq mu^2T$. Its mean is generally biased, so midpoint action uses $2m\mathbb{E}[Y^2]/T$, rather than variance alone.

6. Consequence for the action-gap programme

Finite speed, velocity memory and exact cut consistency coexist with vanishing polygon error and arbitrarily small action scales. Their combination provides a concrete classical admissible family against which a proposed selection principle can be tested. A positive universal remainder requires an additional restriction on this family, traced to its physical role.

The next test is force control: replace impulses by bounded-acceleration turns and determine the minimum duration of a return with prescribed opposite endpoint velocities. This introduces an energy/force timescale through a mechanical constraint, making its action dependence and scaling testable.

Claims and source coverage are maintained separately in the ledger and the B14 audit. The present path law is a specialization of established telegraph probability; the cut/action consequences are derived here for the project test.